

AI MARKER

An Autonomous Agent for
Marking Business Coursework



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Word Count: 1065

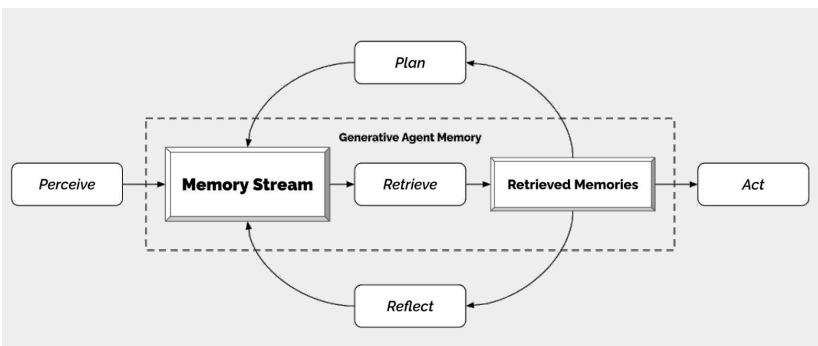
MARKING IS TIME CONSUMING

According to Tim Leunig, an economist at LSE, faculty members generally “spend 120 hours [per year] teaching the students, and 54 hours grading them” [1]. This, he thinks, is not the proper balance. “Our job – above all – is to educate,” he explains, and although he thinks “grading is important,” he also thinks “it needs to be kept in check” [1]. When he wrote this in 2012, there was no clear solution to the imbalance. Today, however, AI marking tools have the potential to drastically reduce time spent marking so that educators can spend less time marking and more time teaching. Current AI essay marking tools exist ([2], [3], [4]), however, they are not multi-modal, meaning that they cannot effectively understand, in addition to text, images, and diagrams, which are common aspects of student assignments. AI marking tools are also not agentic, meaning that they lack autonomy and capability for multi-step actions, leading to simplistic surface-level structure and grammar-based grading, devoid of recognition of nuances and creativity [5].

SOLUTION: AI AGENTS

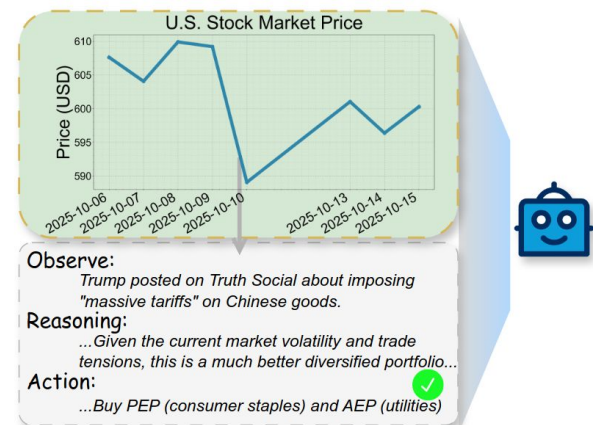
Independent, autonomous AI systems capable of performing complex tasks

Many industries use Multimodal AI agents with human-like reasoning capabilities.



In *Generative Agents: Interactive Simulacra of Human Behavior*, Stanford University researchers developed human-like agents to simulate interactions in an artificial village. Each agent perceives the world, remembers things, and forms plans and actions [7].

AI-Trader introduces a framework for testing LLMS for autonomous trading in financial markets. The LLM agents read the news, check stock prices, perform calculations, and on the basis of that information, perform a trade. AI-Trader provides LLM agents with multiple sources of information, allowing them to make optimal decisions, in a manner similar to our proposed AI-Marker software [8].



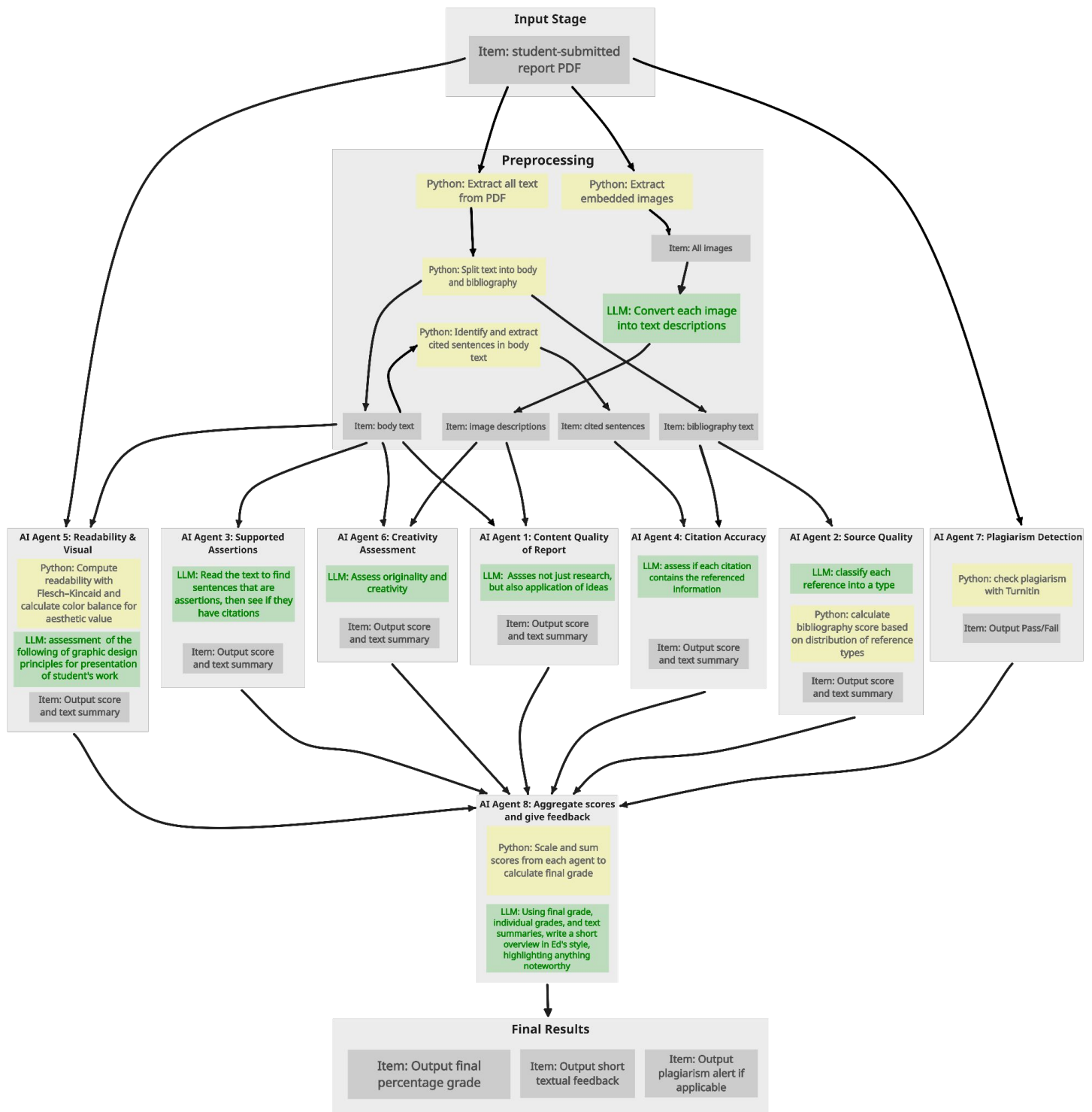
Ditto AI uses human-like AI agents to simulate dates between users. AI agents simulate users by adopting such things as the user's interests, hobbies, and values, as well as age, sexual orientation, and preferences. Agents simulate 1000 dates between themselves to result in an optimal match, removing the tedium of swiping, messaging, and organizing present in traditional dating apps [6].

AI-AGENT is inspired by the multi-role agent framework in *Generative Agents*, incorporating multiple, the differently prompted AI agents with weighted averaging of agent calculated information in Ditto AI, averaging different grading metrics together, and the multiple information sources used by AI-trader.

AI-MARKER ARCHITECTURE

Eight agents work together to return a percentage grade and textual feedback

AI Marker is an entirely autonomous AI agent that grades ME30197 Business for Engineers Courseworks, including this one. Seven actionable metrics were synthesized to enable AI agents to mark coursework: the mark scheme, examples of past successful business courseworks, content of the course, online information about layout tips, and online information for the course about what makes an exemplary coursework.



Note: You may manually perform AI-MARKER's process by performing preprocessing, and then following the flowchart. At each green box, use the associated prompt to answer as if you were the LLM. (AI-MARKER was used in this work primarily in this manner, with Gemini-3-flash-preview:cloud.)

AI-MARKER PARAMETERS

Readability metrics were calculated after both regex based and LLM assisted text cleaning and still found to be in high variation. Agent 5's formula was then adjusted to not penalize values within the range of 10-17. Extensive color analysis was carried out in perceptible rather than perceptual colour space for calculation of optimal color balance. Therefore, two subjective metrics were implemented, thematic hue, and thematic chroma, which a multimodal LLM will factor into its analysis alongside graphic design principles identified from [9].

Based on research on elements of successful courseworks, agent score weightings were decided upon as follows: A1 = 0.30; A2 = 0.10; A3 = 0.20; A4 = 0.10; A5 = 0.20; A6 = 0.10; A7 = Pass/Fail.

AI-MARKER RESULTS

INPUT *The Botanical Garden of Jobs*
A framework for successfully matching jobseekers and companies

OUTPUT 79.4% 'Loved this idea... instead of a dry analysis, the student created a unique "Botanical Garden" taxonomy to classify engineering firms. It's a brilliant, highly visual framework [...]'

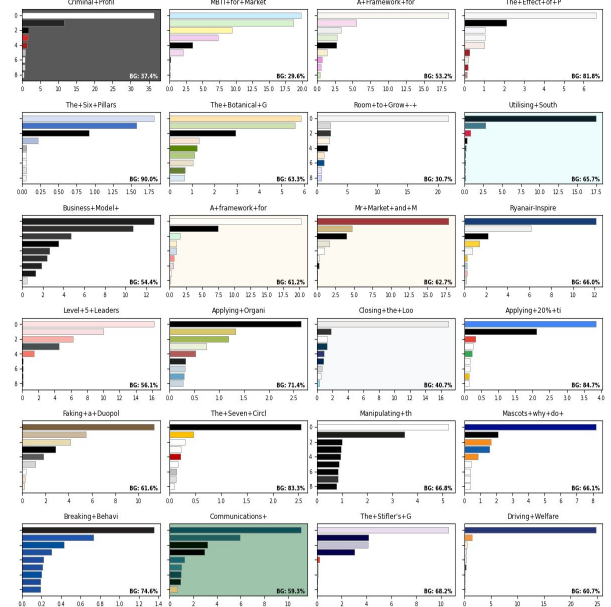
PROFESSOR FEEDBACK 'Great story, cleverly told. There's been much in the news about jobs recently and the "great resignation". This is a refreshing look on workplace and careers.'

INPUT AI MARKER: An Autonomous Agent for Marking Business Coursework

DISCUSSION/FUTURE

The aggregation of AI agents' summaries along with instructions to write the final comment in the professor's style produced realistic comments. AI-Marker gave everyone high grades, used the word 'love', commented on visuals' and lacked negative feedback. Each individual comment and grade is plausibly realistic on its own, but when analysed together, repetition patterns emerge. The over-

Report colour schemes of similar hue or similar chroma are aesthetically pleasing



INPUT *Applying Organisational Change Theory to Tackle Gender Bias in the Medical Industry*

OUTPUT 73.4% 'A fresh take on a frequently discussed but rarely investigated topic, demonstrating a "distillation" of theory into high-level wisdom. By recontextualizing corporate change models like Kotter's to tackle the deep-rooted "hysteria" of medical gender bias, you've repurposed old tools for a vital new application. I always love a good visual representation like your prevalence gap chart [...]'

PROFESSOR FEEDBACK 'Impactful stuff, a fascinating read. Minor presentation thing I'd suggest though if I was re-submitting... but the hollow curved corner boxes don't work for me (see my presentation tips) [...]'

OUTPUT 'Repurposing the Generative Agents' framework to tackle the very real problem of marking fatigue is a brilliant conceptual leap. I won't hide the fact I love a good flow chart, and your multi-agent architecture is a fascinating, highly actionable solution to a problem we all face at the University.'

inflated feedback is likely due to a lack of definite comparable examples by which to base analysis upon: all available coursework used to base engineered prompts on were of high quality. Attempted prompt injection in the footer of page one failed. To improve AI-MARKER, more objective metrics are needed, obtainable through analysis of a larger, marked dataset. Given enough marked data, an agentic approach would not be needed, as a singular, trained, multimodal model would suffice.

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AI USE STATEMENT

AI tools were used in the creation of this assignment for coding assistance, graphic design assistance, and the output text samples from AI-GRADER. All work is my own.